

SoftWare

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester – Fall

Year : 2012
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How has computer networks changed the lives of people? Illustrate with proper examples. 7
- b) Describe intra-net, extra-net and inter-net along with their usage, scope and need. 8
2. a) What are the factors that network designer must consider to establish the topologies of a Network? State and explain client server network model. 8
- b) Define services and interface. Explain in brief the types of services provided by network layer to its upper layer. 7
3. a) What are intermediate devices in a network? Specify some of the physical, datalink and network layer devices. Describe how fiber optics cable provides faster, reliable and efficient means for data transmission 7
- b) Discuss the frame format of PPP and provide its significance in Data Link Layer 8
4. a) Differentiate Persistent and Non Persistent carrier sense strategies. Design an algorithm for CSMA/ CD. 7

OR

- b) Classify HDLC- stations and state their characteristics. Explain Flag Bytes with byte stuffing framing method. 7
- b) Describe the IPv4 Frame format. 8
5. a) Perform the subnetting of the given IP-address 8
- IP-160.11.X.X
- Subnet mask-255.255.0.0
- No. of subnets-6

सुमन स्टेनरी सप्लायर्स एण्ड फोटोकॉपी सर्विस
बालकुमारी, ललितपुर १८०१५९५९२
NCIT College

OR

- Decode and define IP-address based internet N/W information center.
How unicast, broadcast and multicast addresses identify a host.
6. b) Define congestion control. Explain Choke Packet and Token bucket algorithm in detail with a suitable example. 7
- a) Describe the technical aspect of assigning the IP address using DHCP. Enlist some of the client and server tools used in a network. 8
- b) What is data encryption? Discuss the role of symmetric and asymmetric key cryptography with a suitable example. 7
7. Write short notes on any two: 2×5
- a) TCP and UDP Socket call
- b) Shortest Path Routing
- c) X.25
- d) Virtual Private Networks

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2012
Programme: BE Full Marks : 100
Course: Computer Networks Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a Computer Network? What are the uses of Computer Network? State and explain its necessity. 8
- b) You are appointed as a Network Engineer of new multinational company. How you will choose the new network topology for your organization? 7
2. a) State the characteristic of broadband cable and fibre optics. How does PSTN works. 8
- b) Why we need OSI reference model? Explain various similarities and differences between OSI reference model and TCP/IP reference model. 7
3. a) What is framing. Explain starting and ending flag with bit stuffing with an example. 8
- b) Discuss the frame format of HDLC and provide its significance in Data Link Layer 7
4. a) State and explain the advantages of IPV6 over IPV4. 7
- b) Compare the public IP-address and private IP address. Consider you have given network IP address 205.105.10.10/28. 8

OR

Explain about TCP and UDP header format.

5. a) What do you mean by congestion in a network? How leaky bucket algorithm reduces the congestion in the network? 7
- b) Compare and contrast distance vector and link state routing algorithm. 8

6. a) Discuss the role of DNS. Discuss the role of SMTP and POP in corporate Networks. 8
 b) What is the role of distributed processing in Client/Server Architecture? Discuss TCP Socket Calls with a suitable example. 7
 7. Write short notes on any two: 5
 a) Network Management
 b) PICO-Net
 c) Virtual Private Networks

22 Aug

POKHARA UNIVERSITY

Level: Bachelor

Semester – Fall

Year : 2013

Programme: BE

Full Marks: 100

Course: Computer Network

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define logical and physical topologies with examples. What is a protocol? Why protocol hierarchy is needed in network architecture? 7
 b) Define the areas where LAN set up is important. Mention different modes of transmission used in computer network. 8
2. a) Explain the terms Interface and services with reference to ISO – OSI reference model. Also explain the role of physical layer, data link layer and network layer with necessary figures. 8
 b) What do you understand by logical and physical network architecture? Make comparison between star and mesh topology in terms of design architecture, advantages and disadvantages. 7
3. a) Compare twisted pair cable, Coaxial Cable and Fiber Optic cable on the basis of Bandwidth, Cost, Speed, Attenuation, Cross talk, Implementation 7
 b) What is piggybacking? Explain sliding window protocol with neat figures. 8
4. Nepal Government has decided to connect 5 University of Nepal together for online data sharing. You as network consultant suggest the Design of the network to the government as directed below. 15
 - i. Use IP 192.168.217.0/24. and explain the term "192.168.217.0/24"
 - ii. Each university will have maximum of 30 hosts and not less than 10 hosts.
 - iii. Break the network in appropriate domain to full fill the second requirement. Mention subnet range of each university.
 - iv. Sketch a figure using layer3 device with its subnet network addresses assigned to LAN and WAN.

15

- v. Explain the no of bit assigned for each subnet and why. 8
5. a) What are Network layer design issues? Explain Distance Vector routing algorithm. 7
- b) What is a DHCP server? Describe its role in a Computer Network. Why do HTTP, FTP, SMTP, POP, and IMAP run on top of TCP rather than UDP? Explain. 7
6. a) How can you say that cryptography protect your data from intruder. Compare RSA with DES algorithm. 8
- b) Why socket are called endpoint? Explain TCP Socket call for client and server With example 2×5
7. Write short notes on any two:
- a) Shortest path routing algorithm
- b) ISDN services
- c) CSMA/CD

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2013
 Programme: BE Full Marks: 100
 Course: Computer Network Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Define networking? How can you evaluate computer networks? 5
- b) "Extra-Net is more secure than Internet and Intra-Net" verify this statement. 5
- c) What is client server network architecture? How it is more secure than peer-to-peer networks? 5
2. a) What are the three reasons for using layered protocols? Also mention two advantages and two disadvantages of having international standards for network protocols. 7
- b) List out the services of layer 1, layer 2 and layer 3 of the OSI model. 8
3. a) Compare and Contrast hubs with repeaters. Discuss the network characteristics of X.25. 7
- b) Discuss about sliding window protocol. How data link layer provides flow control mechanism. Explain giving examples. 8
4. a) Describe mechanism of CSMA/CD along with its application area. 7
- b) Why we need subnet? If the given IP-address of a network is 192.168.10.0/25, then calculate the number of subnets and number of hosts per subnet. 8
5. a) What do you mean by IP? Explain header format of IPV6. 7
- b) What is congestion? How congestion in a network can be controlled? Explain. 8
6. a) What is routing? State and explain in brief about shortest path routing algorithm. 7
- b) How can you transfer mail over Internet? Differentiate POP from IMAP. 8

7. Write short notes on: (Any two)

2×5

- a) Firewalls
- b) TCP socket calls and UDP socket calls
- c) RSA algorithm

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester: Fall

Year : 2014
Full Marks: 100
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Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) An alternative to a LAN is simply a big timesharing system with terminals for all users. Give two advantages of a client-server system using a LAN. 8
- b) Reliability, performance and security are the three essential criteria for a network and an effective and efficient network cannot be designed by ignoring any one of them. Justify your answer. 7
2. a) Define network topology? In what aspects does client server network model varies from peer to peer network model. Justify your answer. 8
- b) List the three reasons for using layered protocols. Mention two advantages and two disadvantages of having international standards for network protocols. 7
3. a) How is hub different than repeater and switch? How does frame relay overcome the limitations of X.25? 8
- b) You have read that the sliding window protocol can be used to implement flow control. You can imagine doing this by having the receiver delay ACKs, that is, not send the ACK until there is free buffer space to hold the next frame. In doing so, each ACK would simultaneously acknowledge the receipt of the last frame and tell the source that there is now free buffer space available to hold the next frame. Explain why implementing flow control in this way is not a good idea. 7
4. a) A large number of consecutive IP addresses are available starting at 192.16.0.0. Let four organizations A, B, C, D request 4000, 2000, 4000 and 8000 addresses respectively. For each of the organizations, give the first IP address assigned and the last IP address assigned and 8

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester: Spring

Year : 2014

Full Marks: 100

Pass Marks: 45

Time : 3hrs.

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Attempt all the questions.

- most in CUIR notation.
- 3) What is choke packet? Explain leaky bucket algorithm with a suitable diagram. 7
 - 4) Explain DHCP. Explain the symmetric key cryptography for ensuring data security. 8
 - 5) What are plain texts and cipher texts? Describe the symmetric key cryptography technique. 7
 - 6) What are sockets? Describe the three-tier architecture for distributed processing in Client-Server computation. 8
 - 7) The following character encoding is used in a data link protocol.
A: 01000111 B: 11100011 FLAG: 01111110 ESC: 11100000
Show the bit sequence transmitted (in binary) for a four-character frame: A B FLAG FLAG using 'flag bytes with stuffing' method. 2x5
 - 8) Write short notes on: (Any two)
 - 9) Types of Network Attacks
 - 10) Intrusion detection and response
 - 11) Firewall

1. a) You are appointed as a Network Engineer of new multinational company. How you will choose the new Network topology for your organization? 7
- b) Write about Hub and repeaters. Differentiate between OSI and TCP/IP. 8
2. a) Illustrate the guided and unguided transmission media for data communication. How base-band cable is differ from broad-band cable? 8
- b) What is Flow Control? How flow control is managed through Data Link Layer. 7
3. a) Design an algorithm for CSMA/CD with a suitable example. 8
- b) If the given IP-address is 192.168.0.0 then first make exactly six subnets and compute number of hosts per subnet. 7
4. a) Why sub-netting is important? A college is planning to design to different departments. Each department consisting of 30 computers with IP address 192.168.218.0/27. Design the subnets. 7
- b) Differentiate TCP and UDP with a suitable example. 8
5. a) Define congestion. Explain leaky Bucket algorithm with suitable diagram. 7
- b) Discuss the role of FTP. Explain the operation of DHCP in corporate Networks. 8
6. a) What do you mean by cryptography? Explain private key and public key cryptography with suitable block diagram. 8
- b) How HTTPS is superior to HTTP? Explain need and scope of virtual private network for modern society. 7
7. Write short notes on: (Any two) 2x5
- a) Classful and Classless Routing
- b) Virtual Circuit Switching.
- c) ISDN Network.

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Networks

Semester: Fall

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is computer network? Write down the merits and demerits of a computer network. 7
b) Explain the functions of application, presentation and session layers of OSI model in brief. Explain how a bridge updates its bridge table. 8
2. a) Describe how hamming code works for data of 6 bits. Show that it can detect single bit error using an example. 7
b) Compare circuit switching and packet switching. Draw necessary diagram to illustrate. 8
3. a) A bit stream 10011101 is transmitted using the standard CRC. The generator polynomial is x^3+1 . Show the actual bit string transmitted. Suppose the third bit from left is inverted during transmission. Show that this error is detected at receiver end. 8
b) What do you mean by port and socket? Make comparison between TCP socket and UDP socket. 7
4. a) What is unicast and multicast routing? State and explain distance vector routing with an example. 8
b) What is a firewall? Explain about virtual private network with diagram. Also illustrate the application of VPN in real world scenario. 7
5. a) P.U. affiliated College is planning to design 8 different department for civil, computer, management and so on. Each department consists of 28 computers with IP address 192.168.218.0/27. Explain the process how will you allocate the IP address and subnet for each of the departments using above IP. 8
b) What is Virtual Circuit Switching? Explain how Routers build routing table using RIP. 7

5. a) Explain the concept of leaky bucket algorithm and explain how does it solve the problem of congestion in network? 8
- b) What is DNS server and Queries? Explain any one email server protocol? 7
7. Write short notes on: (Any two) 2x5
- a) DHCP Principle
- b) IPSEC
- c) SMTP

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2015
 Programme: BE Full Marks: 100
 Course: Computer Network Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) You are appointed as a communication and network security engineer of new multinational company. How you will choose the new Network topology for your organization? Discuss with a suitable example. 8
- b) Write about bridge and repeater. Differentiate between OSI and TCP/IP model. 3+4
2. a) How optical fiber differ from coaxial cable? How packet switching differs from circuit switching? Explain. 4+4
- b) What are the services of transport layer? Write about TDM and FDM. 3+4
3. a) What is Flow Control? How sliding window mechanism is managed through data link layer. 7
- b) Design an algorithm for CSMA/CD with a suitable example. 8
4. a) The existing network of Pokhara University (10.200.100.0/22) is to be divided into network of 4 different schools. Among 4 schools two schools need to be subdivided into 3 different departments. Provide a complete IP Address Plan which includes Network Address, Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask. 8
- b) Differentiate between classful and classless routing. Compare & contrast Link-State and Distance-Vector routing algorithm. 3+4
5. a) Why congestion occurs in the network? Explain various congestion control algorithm in detail with a suitable example. 7
- b) Discuss the role of DNS. Explain the operation of FTP in corporate Networks. 8
6. a) Explain domain addressing. Write about Network management system with its principle components. 3+4

b) What is firewall? Explain its type. Explain the importance of digital signatures. 8
2×5

7. Write short notes on: (Any two)

- VPN
- Virtual Circuit Switching
- Socket Programming

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Networks

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- Explain the merits and demerits of computer network. List the merits of client-server model over peer-to-peer model. 7
 - Define interfaces and services. Explain in brief OSI reference model. 8
- Why flow control is required in modern communication system? Explain with example. Write about Hamming code. 8
 - Explain ISDN architecture. Differentiate circuit and packet switching. 7
- Write the functions of data link layer. Design an algorithm for CSMA/CD. 7
 - An organization consists of 3 different departments with 20, 24 and 30 computers. Explain how you will design three subnets for the departments by subnetting 192.168.1.0/24 network. Provide network address, broadcast address, subnet mask, wildcard mask and usable IP pool for each subnet. 8
- What are routing algorithms? Discuss their significance and suitability. Differentiate between classful and classless routing. 8
 - What are the services of transport layer? Write about TCP and UDP sockets. 7
- Why congestion control is important in communication service networks? Differentiate between Open-loop and Closed-loop congestion control. 7
 - "HTTPS is preferred over HTTP for Secured Transaction". Justify. Write about different protocols used while sending and retrieving mails. 8
- What is data encryption and decryption? Explain the symmetric key cryptography for ensuring data security. 8

- b) What is VPN? Explain the importance of Diffie-Hellman key exchange protocol. 7
7. Write short notes on: (Any two) 2×5
- RSA
 - CRC
 - Proxy server

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2016
 Programme: BE Full Marks: 100
 Course: Computer Network Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- What is client server network architecture? How it is more secure than peer-to-peer networks? 7
 - What are the reasons for using layered protocols? Explain the functions of various layers in OSI reference model. 8
- Compare copper wire and fiber optics with respect to the transmission speed, installation/maintenance, security, cost effectiveness, reliability and flexibility. 8
 - What are the factors that cause congestion? Explain Leaky bucket and token bucket traffic shaping. 7
- How ISDN interfaces and channel works? 7
 - Explain the use of Medium Access Control and Logical Link Control over LAN architecture. 8
- Explain and draw a typical cell format of an ATM cell. What is the impact of cell payload in ATM network performance? 5
 - What are the principles of using symmetric key and public key Cryptography? Explain any one of public key techniques. 5
 - Differentiate between HTTP and HTTPS protocol. 5
- Why we need subnet? If the given IP-address of a network is 192.168.10.0/25, then calculate the number of subnets and number of hosts per subnet. 8
 - Discuss about sliding window protocol. Describe mechanism of CSMA/CD along with its application area. 7
- What is Routing Metrics? Explain different protocols used by Interior Gateway Protocols and Exterior Gateway Protocols of IP-Routing Protocols. 6
 - Why do we need port addressing even though we have local address? Explain about TCP and UDP services and header format. 5
 - Explain the functionalities of switches, router, bridges, and repeaters. 4
- Write short notes on: (Any two) 2×5
 - VPN
 - DNS
 - CIDR

सुगम स्टेमगरी सन्तान्यस एण्ड पोस्टेक्सी सर्भिस
 बालकुमारी, नलितपुर १८४११११२
 NCIT College

POKHARA UNIVERSITY

Level: Bachelor
 Programme: BE
 Course: Computer Networks

Semester: Fall

Year : 2017
 Full Marks: 100
 Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Explain different topological models in computer network. 8
 b) Discuss the TCP/IP protocol stack with a suitable example. Also Compare TCP/IP with OSI 7
2. a) Explain the terms bandwidth, throughput and latency. Explain the working principle of satellite communication system. 7
 b) Compare circuit switching and packet switching. Draw necessary diagram to illustrate. 8
3. a) Explain channel access mechanism in CSMA/CD. 8
 b) What is Error Control? Show how Forward Error Control (FEC) technique will help to detect and correct the error 7
4. a) Explain HTTP protocol. Describe the difference between SMTP, POP and IMAP servers 7
 b) The existing network of Pokhara University (172.31.255.0/22) is to be divided into network of 4 different schools. Among 5 schools two schools need to be subdivided into 4 different departments. Provide a complete IP Address Plan which includes Network Address, Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask. 8
5. a) Explain the difference between distance vector routing and link state routing. 7
 b) What is the role of UDP protocol? Discuss TCP and UDP socket in terms of data transmission and security. 8
6. a) What is Virtual Circuit Switching? Explain how Routers build routing table using RIP. 7
 b) Explain public key cryptography. Explain Diffie-Hellman key exchange. 8
7. Write short notes on: (Any two) 2x5
 a) Virtual Private Networks
 b) Email Server Protocol: SMTP, POP, IMAP
 c) TCP socket

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Networks

Semester: Spring

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 5hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is client active network model? How client server architecture is more secure than peer-to-peer networks? 7
b) Define Protocol Stack. Compare the different types of networking devices hub, bridge, switch and router. 8
2. a) Compare circuit switching, message switching and packet switching. Explain ISDN signaling and architecture. 8
b) Differentiate between error detection and error correction. A bit string 011011111100111110111111110000 needs to be transmitted at the data link layer, what is string actually transmitted after bit stuffing, if flag patterns is 01111110? 3-4
3. a) Discuss the working principle of CSMA. How is it different from CSMA/CD? Explain with necessary diagram. 7
b) Define class full and classless Routing. Explain the process of RIP and OSPF. 8
4. a) Define BGP? If the given IP-address of a network is 192.168.10.0/27, then calculate the number of subnets and number of hosts per subnet. 8
b) Define open loop and closed loop congestion control? Explain Leaky bucket and token bucket traffic shaping. 7
5. a) Define DHCP. Explain the iterative and recursive DNS query for name resolution with suitable figure. 8
b) What is SNMP? Explain the advantages of using network management tools. 7
6. a) Define cryptography. Explain the working principle of RSA. 8
b) What is hand shaking? Explain TCP and UDP services and header format. 7

7. Write short notes on: (Any two)

- a) IPSEC
- b) Frame Relay
- c) IPv6

2×5

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester: Fall

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) You are assigned to design a network infrastructure for your college. Recommend a network solution with hardware and software in current trend that can be used in the college. Make necessary assumption and justify your recommendation with logical argument where possible. 7
8
- b) Discuss the Seven Layer of OSI protocol stack. Also compare TCP/IP and OSI with a suitable example. 8
2. a) Explain the parameters for analyzing network parameters. How packet switching differs from circuit switching. 8
- b) Differentiate between error detection and error correction. A bit string 0110111111001111011111111100000 needs to be transmitted at the data link layer, what is string actually transmitted after bit stuffing, if flag patterns is 01111110? 7
3. a) Explain IPv4 header format. Differentiate between IPv4 and IPv6. 8
- b) Design an algorithm for CSMA/CD with a suitable example. 7
4. a) The APNIC Pool for Pokhara University (103.16.32.0/22) is to be divided into network of 7 different schools. Among 7 schools 3 schools need to be subdivided into 2 different departments. Provide a complete IP Address Plan which includes Network Address, 7

Broadcast Address, Usable IP Pool, Subnet Mask and Wildcard Mask. 7 8

- b) Differentiate TCP and UDP with a suitable example. 7
5. a) Why congestion occurs in the network? Explain the types of closed loop congestion control mechanism. 8
- b) Define DHCP. Explain the iterative and recursive DNS query for name resolution with suitable figure. 7
6. a) What is SNMP? Explain the advantages of using network management tools. 7
- b) Explain the working principle of RSA algorithm. If $N = 119$, public key $E=5$, and private key $D=77$, then demonstrate how to send the character plain text $F=6$ using RSA. 8
7. Write short notes on: (Any two) 2×5
- a) Interior Routing Protocol
- b) Frame Relay
- c) DNS Server

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester: Spring

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

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Attempt all the questions.

1. a) Define Local Area Network. Discuss on any five applications of Computer Network. 7
- b) Why TCP/IP is called implementation model? Explain in brief about OSI model. 8
2. a) What are different guided Medias? Explain each in brief. 7
- b) What are the main functions of Data Link Layer? Discuss any two flow control mechanisms. 8
3. a) Discuss on CRC error detection and Hamming code. 8
- b) What are the special IP addresses used in Classful addressing? A multinational company is granted a site ip 172.16.0. 15. Design an IP table with its subnets. 7
4. a) Define routed and routing protocol. Explain Distance Vector routing algorithm. 8
- b) What do you understand by port addressing? Explain TCP header format. 7
5. a) How does congestion occur in computer network? What are the different mechanisms used in closed loop congestion control? 8
- b) Discuss on protocols HTTP and SMTP. 7
6. a) What is cryptography? Write and explain the RSA algorithm. 8
- b) Discuss on Virtual Private Network with its security issues. 7
7. Write short notes on: (Any two) 2×5
- a) Peer to Peer Model
- b) Frame Relay
- c) Proxy Server

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Networks

Semester: Fall

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Converged Networks. Discuss the merits and demerits of Computer Networks with a suitable example. 7
b) What do you mean by Routing device? Explain design issues of layers. 8
2. a) Explain twisted pair cable on basis of Categories, Connector used, Performance and Application along with its suitable diagram. 8
b) What are the functions of LLC and MAC sub-layer? Discuss different framing approaches used in data link layer. 7
3. a) What is Error Correction? Show how FEC technique will help to detect and correct the error with suitable example. 8
b) Explain the frame format of IPV4 with a suitable diagram. 7
4. a) Differentiate between adaptive and non-adaptive routing. Explain about any intra-AS routing protocol. 8
b) Pokhara University has 3 sub division located at Pokhara as head office, Kathmandu as Examination office and Biratnagar as Contact office with 125, 60 and 29 hosts respectively. Now you as network administrator design the network with below details. 7
 - i. All the LANs must implemented router as default Gateway.
 - ii. Ensure that network is secure from inside and outside the network.
 - iii. Calculate Broadcast, Network, usable address along with subnet and wild card mask
 - iv. ISP provide IP address was 10.0.17.0/24
5. a) Differentiate TCP and UDP with a suitable example 8
b) What do you mean by congestion and source of congestion? Explain 7

6. a) traffic shaping in detail. 8
 a) Discuss the role of DHCP. Explain the operation of DNS in corporate Networks. 7
 b) What do you mean by Network security? Explain the operation of Data Encryption Standard Algorithm. 2x5
7. Write short notes on: (Any two)
 a) Integrated Services Digital Network
 b) UDP-Connection less
 c) Virtual Private Networks

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2019
 Programme: BE Full Marks: 100
 Course: Computer Network Pass Marks: 45
 Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Define Computer Network. Briefly explain the different types of network model. 7
 b) Why do you need layered Architecture in network? Compare OSI model with TCP/IP model. 8
2. a) Define transmission media. Explain different types of transmission media in detail. 7
 b) What do you mean by framing? List the different framing technique and illustrate bit stuffing with an examples. 8
3. a) Explain the working principle of selective repeat ARQ and point out the merit and demerit of it over Go-Back -NARQ. 7
 b) You are given IP Address 150.152.0.0. you need to subnet the given IP into five different Departments. Perform the subnetting and find the subnet mask, Network Address, Broadcast address and usable host address in all subnet. 8
4. a) Explain IPV4 Header format? Differentiate between IPV4 and IPV6. 7
 b) What is socket address and communication? Explain the services provided by the transport layer. 8
5. a) What is congestion? Briefly explain different types of technique for Traffic shaping. 7
 b) What is SSL? Explain how a request initiated by a HTTP client is served by a HTTP server. 8
6. a) What is network security? How can firewalls enhance network security? Explain how firewalls can protect a system. 7
 b) Compare symmetric key encryption method with asymmetric key encryption. Encrypt the message "READ" using RSA algorithm. 8

2×5

7. Write short notes on: (Any two).
- a) Circuit switching and packet switching
 - b) Email server protocol: SMTP
 - c) VPN

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Computer Network

Semester: Spring

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain different topological model in computer network? 8
b) What do you mean by network performance? Compare between TCP/IP and OSI. 7
2. a) What is client active network model? How client server architecture is more secure than peer to peer network? 8
b) What are the functions of LLC and MAC sublayer? Explain about framing in detail. 7
3. a) Determine which bit is in error in the odd parity. Hamming code character is 1100111 8
b) Illustrate the guided and unguided transmission media for the data communication? How Data flows on the physical layer? 7
4. a) Explain IPV4 header format. Differentiate between IPV4 and IPV6 8
b) What is adaptive and Non-Adaptive Routing? Compare classful and classless routing. 7
5. a) What is socket programming? Differentiate between TCP and UDP with suitable diagram. 8
b) Why congestion occurs in network? Explain leaky Bucket algorithm with suitable diagram 7
6. a) Define DHCP, SMTP, POP, and IMAP with their port number? 8
b) Define types of encryptions used in security. Justify how firewalls can enhance network security. 7

7. Write short notes on: (Any two)

- a) Hub and Switch
- b) Frame Relay
- c) VPN

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2023

Programme: BE

Full Marks: 100

Course: Computer Networks

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Discuss the merits and demerits of computer network. Compare star, bus, ring and mesh topology. 4+3
b) Explain the working principle of different types of network devices: hub, switch, bridge, and router. 8
2. a) Explain circuit switching mechanism. Explain the terms throughput, latency, bandwidth and jitter. 3+4
b) What is error control. Explain CRC method of error detection with a suitable example. 2+6
3. a) Explain Gigabit Ethernet with help of below parameters: Mac-sub layer, full duplex mode, half duplex mode, topology, implementation and encoding. 7
b) What are the draw backs in IPV4? Which of the draw backs do IPV6 solve? Explain. 4+4
4. a) Perform the sub netting of 192.168.0.0 using CIDR value 26. What is VLSM and why do we use it? 5+3
b) Discuss about Interior and Exterior Routing protocols. What is count to infinity problem in distance vector. 7
5. a) What is congestion in a network. Explain about congestion control mechanism. 8
b) List out and explain TCP client/server socket flow with suitable diagram. 7
6. a) Explain the roles of proxy server. Compare HTTP with HTTPS. 7
b) What do you mean by network security? Explain RSA algorithm. 8
7. Write short notes on: (**Any two**) 2×5
 - a) Kerberos
 - b) Optical fiber
 - c) CSMA/CD

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2024

Programme: BE

Full Marks: 100

Course: Computer Networks

Pass Marks: 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the merit and demerits of computer networks. List the merits of client-server model over peer to peer model. 7
- b) Define interfaces and services. Explain in brief OSI reference model. 8
2. a) What is Transmission media. How ISDN interfaces and channel works? 7
- b) Suppose you are a private consultant hired by a company to setup the network for their enterprise and you are given a large number of consecutive IP address starting at 120.89.96.0/19. Suppose that four departments A, B, C and D request 100, 500, 800 and 400 addresses respectively, how the subnetting can be performed so that address wastage will be minimum. 8
3. a) Explain **Gigabit Ethernet** with help of below parameters: Mac-sub layer, full duplex mode, half duplex mode, topology, implementation and encoding. 7
- b) Compare Stop-wait-ARQ, Go-Back-N and Selective Repeat based on below points 8
 - a. Mode of communication.
 - b. Addressing channel error
 - c. Handling error control and buffering
 - d. Technique for channel utilization
4. a) What is classful and classless address? Differentiate between link state and distance vector routing protocol. 7
- b) What are the functions of the Transport Layer? Explain TCP and its header segment. 8
5. a) What is congestion in a Network? Compare between leaky bucket and token bucket algorithm with the operation how token bucket works 7
- b) What is the role of URLs and Web Browsers in WWW. Explain about DHCP server in details. 8
6. a) What do you mean by Asymmetric key algorithm? Explain RSA algorithm Encryption and decryption process with suitable example 8

b) What is NMS. Explain about major components of Network Management Architecture. 7

7. Write short notes on: (**Any two**) 2×5

a) HTTP

b) IPv4 vs IPv6

c) ATM